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SQL QUERY GENERATOR

Natural Language to SQL using IBM Granite 3.3 2B

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ABSTRACT

The SQL Query Generator is an intelligent system designed to convert natural language queries into structured SQL queries. Writing SQL queries manually can be difficult for beginners because it requires knowledge of database syntax and structure.

This project simplifies the process by allowing users to enter queries in plain English. The system analyzes the request and automatically generates the appropriate SQL query.

The application uses the IBM Granite 3.3 2B Instruct model to understand user input and convert it into accurate SQL queries. A user-friendly chat interface built using Gradio allows users to interact with the system easily.

The system also includes security validation mechanisms to prevent SQL injection attacks and malicious commands. Additional features such as exporting queries to PDF and text files are also included.

This project demonstrates how artificial intelligence can simplify database interaction and make SQL accessible to both technical and non-technical users.

1. INTRODUCTION

Databases are widely used to store and manage large amounts of data in modern applications. SQL (Structured Query Language) is the standard language used to retrieve and manipulate data stored in relational databases.

However, many users find SQL difficult to learn because it requires knowledge of query syntax, tables, and database structure. Writing complex SQL queries can be challenging, especially for beginners.

The SQL Query Generator project is designed to solve this problem by converting natural language queries into SQL statements automatically. Users can type their requests in simple English, and the system will generate the corresponding SQL query.

The system uses artificial intelligence techniques to analyze user input and detect the appropriate tables, columns, and conditions required for the query.

By combining natural language processing with database technology, the project simplifies the process of querying databases and improves accessibility for users.

2. OBJECTIVES OF THE PROJECT

The objectives of this project are:

- To convert natural language input into SQL queries automatically
- To reduce the complexity of writing SQL queries manually
- To provide a simple and interactive user interface
- To prevent SQL injection attacks and malicious queries
- To allow users to export generated queries as PDF or text files
- To make database interaction easier for non-technical users

3. TECHNOLOGIES USED

Python

Python is used as the main programming language for implementing the system logic and integrating different modules.

IBM Granite 3.3 2B Instruct Model

This AI model is used for natural language understanding and SQL query generation.

Gradio

Gradio is used to create the web-based chat interface where users can interact with the system.

SQLite

SQLite is used as the relational database schema for generating SQL queries.

Transformers Library

The Hugging Face Transformers library is used to load and run the AI model.

ReportLab

ReportLab is used to generate PDF reports containing the generated SQL queries.

Google Colab

Google Colab provides a cloud-based environment for developing and running the project.

4. SYSTEM ARCHITECTURE

The system architecture consists of multiple modules working together to process user requests and generate SQL queries.

User Interface

Users enter their queries in natural language through the Gradio chat interface.

Natural Language Processing Module

This module analyzes the user input and extracts important keywords such as table names and conditions.

SQL Query Generation Module

This module generates the SQL query based on detected tables, columns, and filtering conditions.

Security Validation Module

This module checks the generated SQL query for dangerous keywords and prevents SQL injection attacks.

Export Module

The export module allows users to download the generated SQL queries as PDF or text files.

5. DATABASE

The system includes four main database tables.

Users Table

Fields:

- id
- name
- email
- signup_date
- age
- country
- status

Orders Table

Fields:

- id
- user_id
- product_name
- amount
- order_date
- status

Products Table

Fields:

- id
- name
- price
- category
- stock

Transactions Table

Fields:

- id
- user_id
- amount
- date
- typ

6. IMPLEMENTATION METHODOLOGY

The project is implemented in several stages.

Environment Setup

Google Colab is used to set up the development environment and install required libraries.

Model Configuration

The IBM Granite model is loaded using the Transformers library.

Query Generation Logic

Functions are implemented to detect database tables, identify columns, and extract filtering conditions from user input.

SQL queries Security Validation

A validation system is implemented to detect dangerous SQL commands and prevent database misuse.

Interface Development

The Gradio framework is used to build an interactive chat interface.

Export Feature

ReportLab is used to generate PDF reports containing.

7. SYSTEM FEATURES

The system provides several features including:

- Natural language query input
- Automatic SQL query generation
- Intelligent table detection
- Security validation for safe query generation
- Export queries as PDF and text files
- User-friendly chat interface

8. TESTING AND RESULTS

Test Case 1

Input:
Show users who signed up last month

Output:

```
SELECT * FROM users
WHERE DATE(signup_date) >= DATE('now','-1 month')
LIMIT 10
```

Test Case 2

Input:
Find transactions greater than 500 from last week

Output:

```
SELECT * FROM transactions
WHERE amount > 500
AND DATE(date) >= DATE('now','-7 days')
LIMIT 10
```

The testing results show that the system successfully converts natural language queries into correct SQL statements.

9. FUTURE SCOPE

The system can be improved by adding the following features:

- Support for complex SQL queries such as JOIN operations
- Integration with larger database systems such as MySQL and PostgreSQL
- Support for aggregation functions such as COUNT, SUM, and AVG
- Allowing users to upload custom database schemas
- Improving the accuracy of query generation

10. CONCLUSION

The SQL Query Generator is an intelligent system that converts natural language queries into SQL statements automatically. By integrating artificial intelligence with database technologies, the system simplifies database querying for users.

The project demonstrates how natural language processing can improve the usability of database systems and reduce the complexity of writing SQL queries manually.

The system provides a simple, secure, and efficient method for generating SQL queries and retrieving information from databases.